

Checkpoint Reporting System - Updated Documentation

Project Overview

This document describes the Siemens PLC -> Node-RED -> MySQL checkpoint logging architecture and the future reporting software requirements.

Current Data Flow

S7 In Node reads checkpoint arrays from DB2. Function 78 adds machine_id. Checkpoint Delta Insert detects changes and writes only changed checkpoints into MySQL using INSERT IGNORE.

Database Structure

Table: checkpoints. Stores machine_id, checkpoint_no, checkpoint_ok, checkpoint_not_ok, time_taken, start_time, end_time and created_at.

Reporting Software Requirements

Dashboard, machine-wise reports, date-range reports, checkpoint performance reports, monthly reports, PDF export, Excel export and audit trail screens.

Machine List Extracted From TPM Workbook

- 8_MD -Abro Balancing(2)
- LINE3 -MIG Welding-2 (2)
- 18_Z101 - Robo Welding (2)
- 22_Z101 - Schenk Balancing (2)
- 6_HD -MIG Welding (2)
- 8_HD -Schenk Balancing (2)
- 8_HD -Abro Balancing(2)
- 11_LD -MIG Welding (2)
- 8_LD -Schenck Balancing(2)
- MD -MIG Welding-1 (2)

Recommended Machine Identification

Use a stable machine_id in Node-RED and store a user-friendly machine_name for reporting. Avoid using demo IDs such as MH_01 in production systems.